Siddhanthaditiyaa Vettakal

Cybersecurity · AI/ML · Full-stack

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BTech Computer Engineering student specialising in Cybersecurity, focused on network defence and attack detection. I build AI-driven security tools and full-stack products end to end, and placed 3rd at the IEEE PCE Cognithon 2026 hackathon. Looking for a cybersecurity internship.

Education

BTech, Computer Engineering (Cybersecurity), Pillai College of Engineering, Panvel — Expected 2028

Experience

Intern, Pillai College of Engineering — Security · AI — Mentor: Dr. Karpagavalli S.
ShelfSense AI · Node.js, MongoDB, YOLOv8, Groq · GitHub · Live

- Built a multi-tenant retail intelligence platform with 40 AI agents (demand forecasting, anomaly detection, churn prediction) and a YOLOv8 detector that finds empty shelf slots in photos.
- Designed a 13-layer security architecture and assessed it against OWASP Top 10, ISO 27001 and NIST CSF. Backend exposes 461 API routes.

Projects

NetForecast: AI-based network attack forecasting (Smart India Hackathon 2026) — Security · AI — GitHub
Python, FastAPI, XGBoost, CICIDS2017, WebSocket

- Built a model that forecasts the next attack type (four DoS variants, DDoS, port scan) from real multi-day CICIDS2017 traffic, using strict time-based splits to prevent data leakage.
- 99.0% accuracy on same-day validation; 87.7% precision and 0.919 ROC-AUC on attack types unseen in training. Live predictions stream to a dashboard over WebSocket.

Pravesh: JEE study coach (IEEE PCE Cognithon 2026) — 3rd place · AI · Security — GitHub
Python, FastAPI, SQLite, JavaScript, RAG, OCR

- Built a study coach that gives a hint first and cites the concept note behind every answer, with a rules-based mastery score and prerequisite-first study plans. Added guardrails against prompt injection and off-topic requests, plus an audit log.

NoteBuddy: AI study companion (college AI competition) — AI agents · Full-stack — GitHub · Live
React, FastAPI, Gemini, Supabase, pgvector

- Built four tool-calling AI agents (study planner, coach, cross-note tutor, organiser), each scoped to the signed-in user's data, plus spaced repetition and semantic search. Deployed on Vercel and Render.

Recoup: travel disruption recovery engine (Hackcelestial) — AI · Full-stack — GitHub · Live
React, Node.js, Express, Gemini

- Modelled trips as dependency graphs and used breadth-first search to find every booking a delay breaks; Gemini writes recovery plans and vendor emails.

EduPulse AI: student performance prediction — ML · Full-stack — GitHub
React, FastAPI, XGBoost, scikit-learn

- Predicts exam outcomes, flags weak topics and generates study plans, with an AI tutor on the Groq API.

Research

- Unpublished paper on ShelfSense AI: its multi-agent retail platform and 13-layer security design.
- Co-authored a research proposal and grant application, "Assessing the Capability of Generative AI in Enhancing Deepfake-Based Social Engineering Attacks" (mentor: Dr. Malavika Sharma).
- Unpublished paper, "Investigating Different Types of Malware Attacks and How These Threats Are Neutralized", including a Likert-scale survey design.

Skills

Cybersecurity	Network security, intrusion detection, malware analysis, OWASP Top 10, secure coding basics, Wireshark, GNS3, OSPF and RIP routing
AI / ML	Building AI agents (tool calling, multi-agent systems), RAG, scikit-learn, XGBoost, YOLOv8, Gemini, Groq, Explainable AI (SHAP)
Coding languages	Python, Java, C, JavaScript, HTML and CSS
Full-stack	React, FastAPI, Node.js and Express, MongoDB, PostgreSQL (Supabase), SQLite, Tailwind CSS
Tools	Git and GitHub, Docker, Vercel, Render

Achievements and languages

- **3rd place**, IEEE PCE Cognithon 2026 hackathon. Participant, Smart India Hackathon 2026 and Hackcelestial.
- **Languages:** English, Hindi, Marathi, Malayalam, German (B2).